# IMPACT CHALLENGE

Problem to Prototype

# AI-Powered Cyber Threat Intelligence




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# Objective

What the Project Aims to Achieve – To develop a simple and reliable Arduino-based smart control system that can automatically operate multiple LEDs according to a programmed sequence.

Main Goal of the IoT Solution – To demonstrate how smart automation and microcontroller technology can be used to control electronic devices efficiently with minimal human intervention.

# Problem Statement

Real-World Application – To demonstrate a concept that can be adapted for smart lighting, traffic management, warning systems, security indicators, and other automated applications.

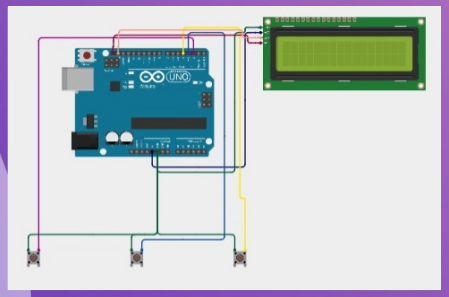
# Proposed Solution & Circuit Diagram

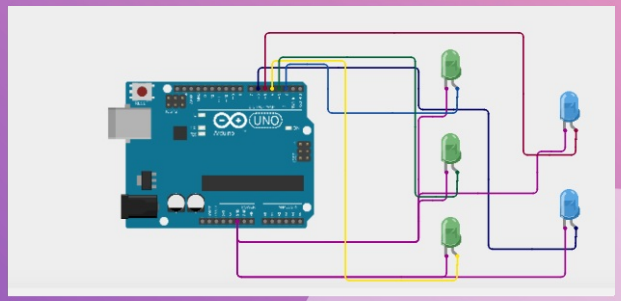
How the Model Works

Three buttons simulate suspicious security events: unusual login, sensitive-file access, and USB data copying.

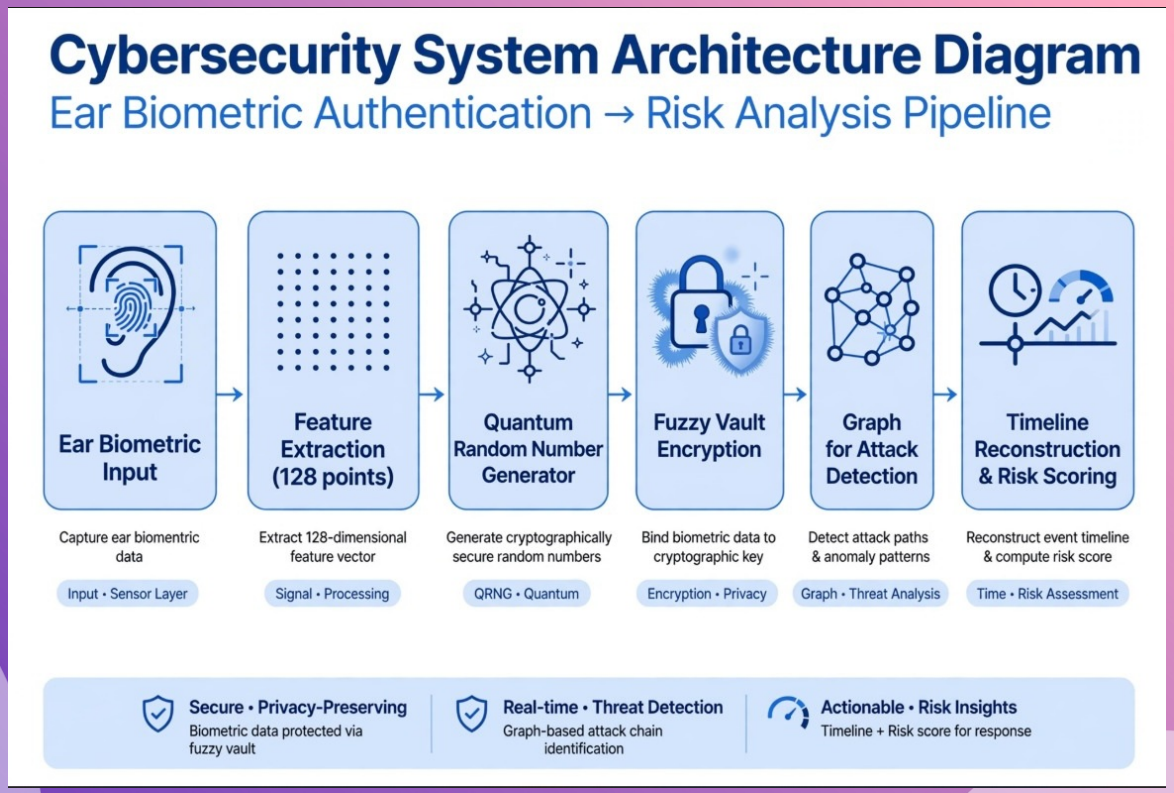
The Arduino records the events and checks whether they are linked to the same user and workstation and occur in the correct order within 10 minutes.

A rules-based risk score is then calculated. If all events are connected, the score reaches 100/100, and the LCD displays the detected incident and recommended actions.





# Future Enhancement

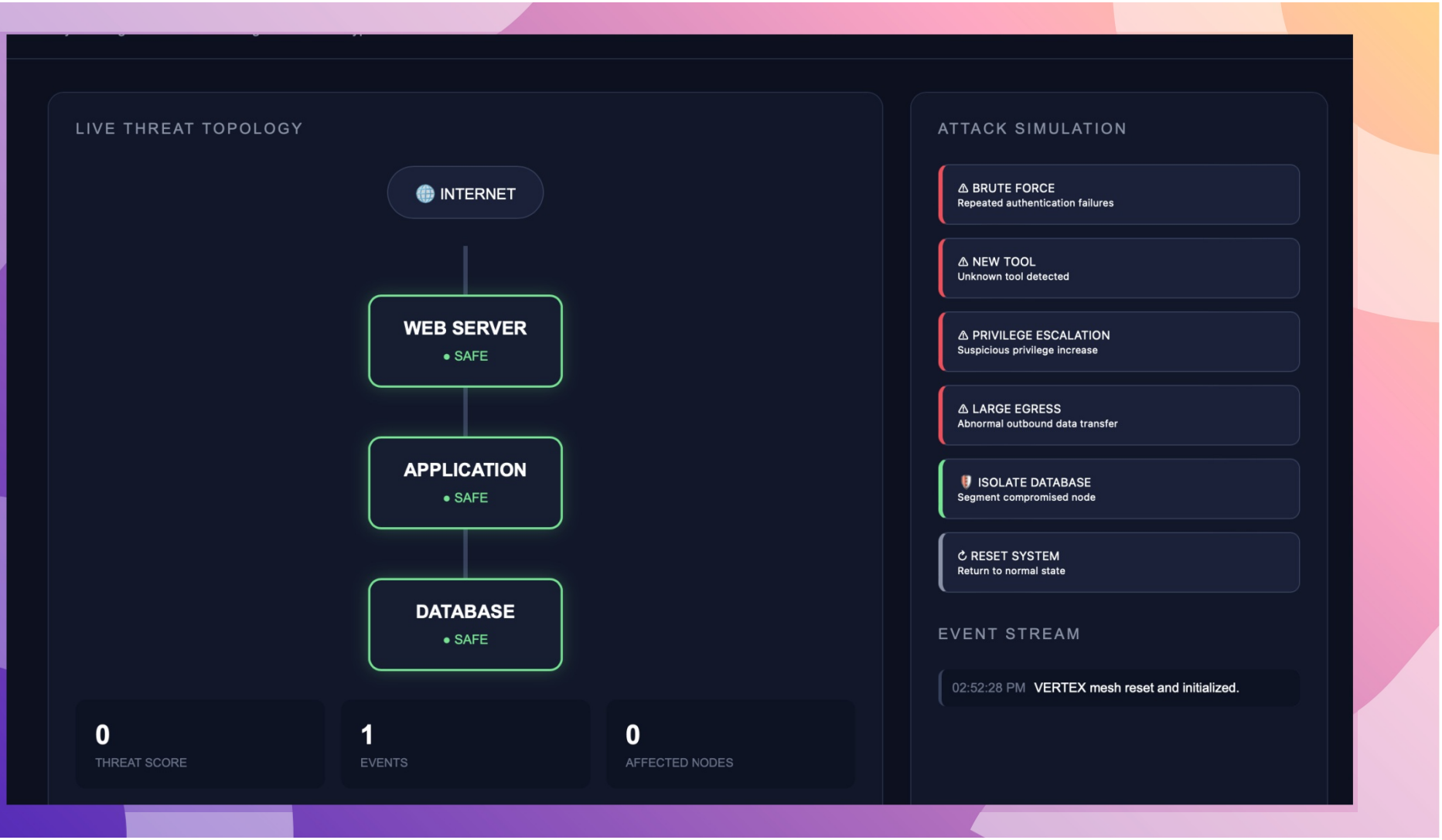


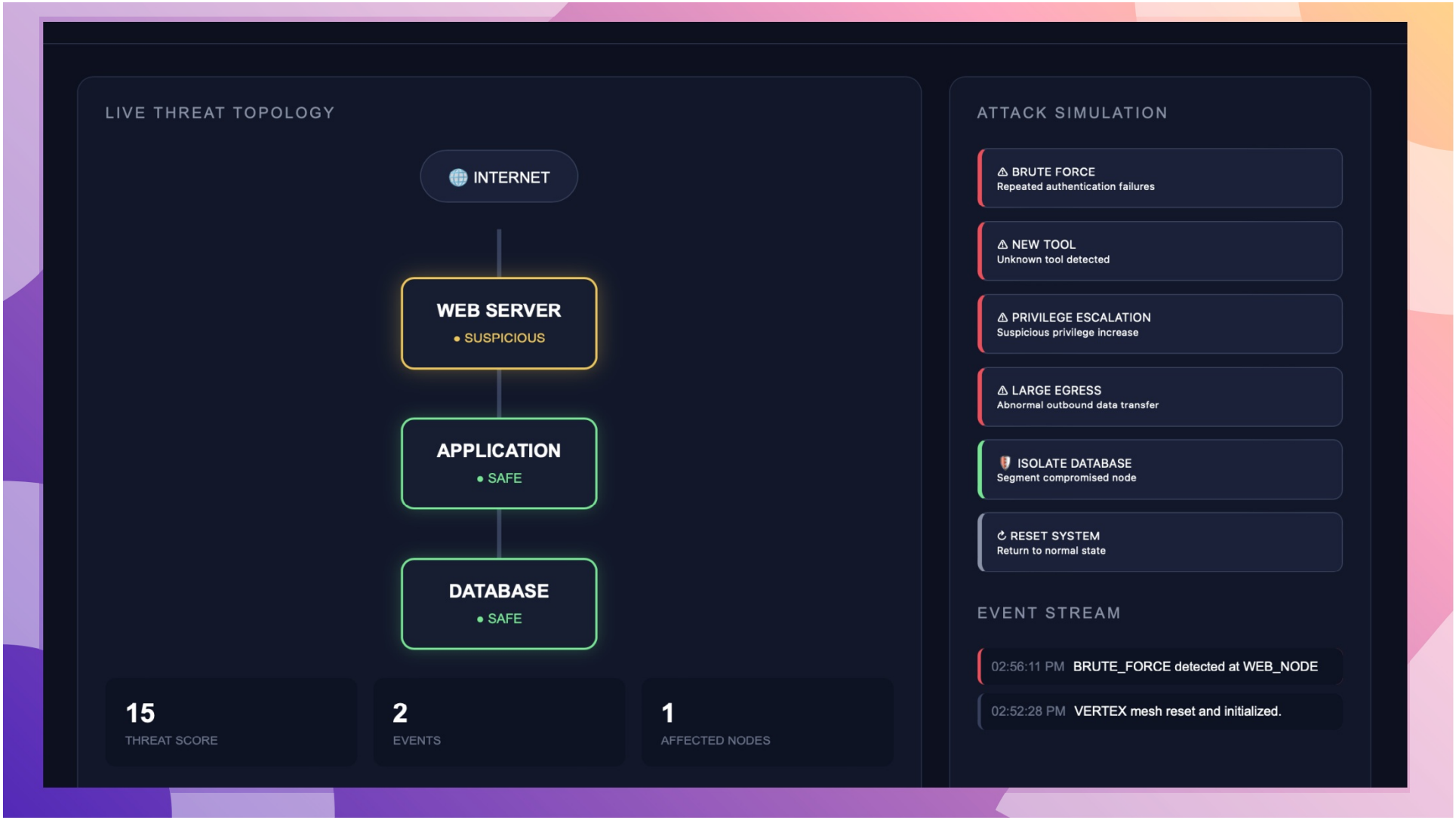
# Conclusion

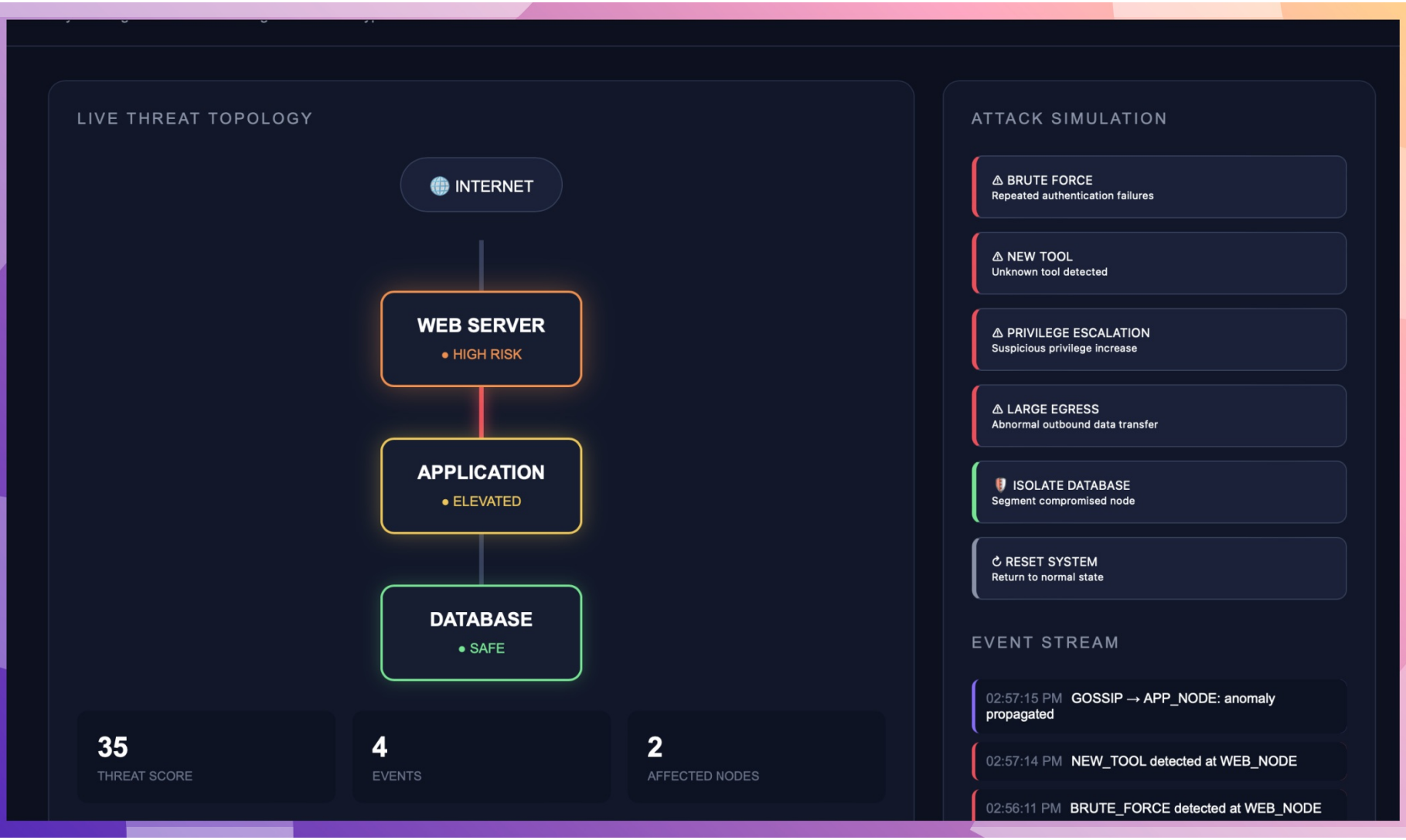
From our study, we found that this type of prototype is useful for understanding the relationship between hardware and software.

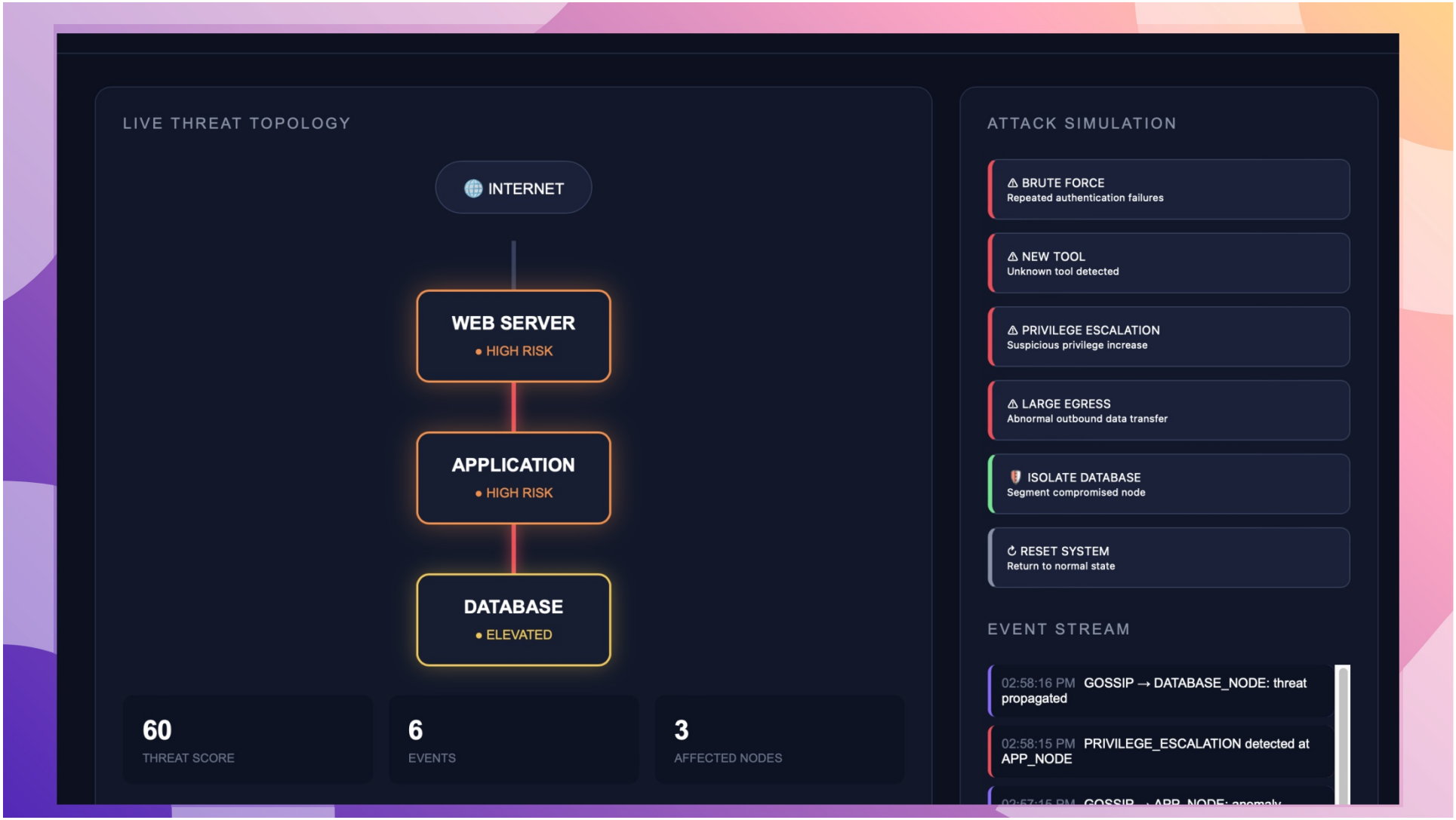
The hardware provides the physical components such as the Arduino, LEDs, wires, and power connections, while the software determines how those components behave.
This makes the model a good example of an embedded system, where a microcontroller is programmed to perform a specific function.

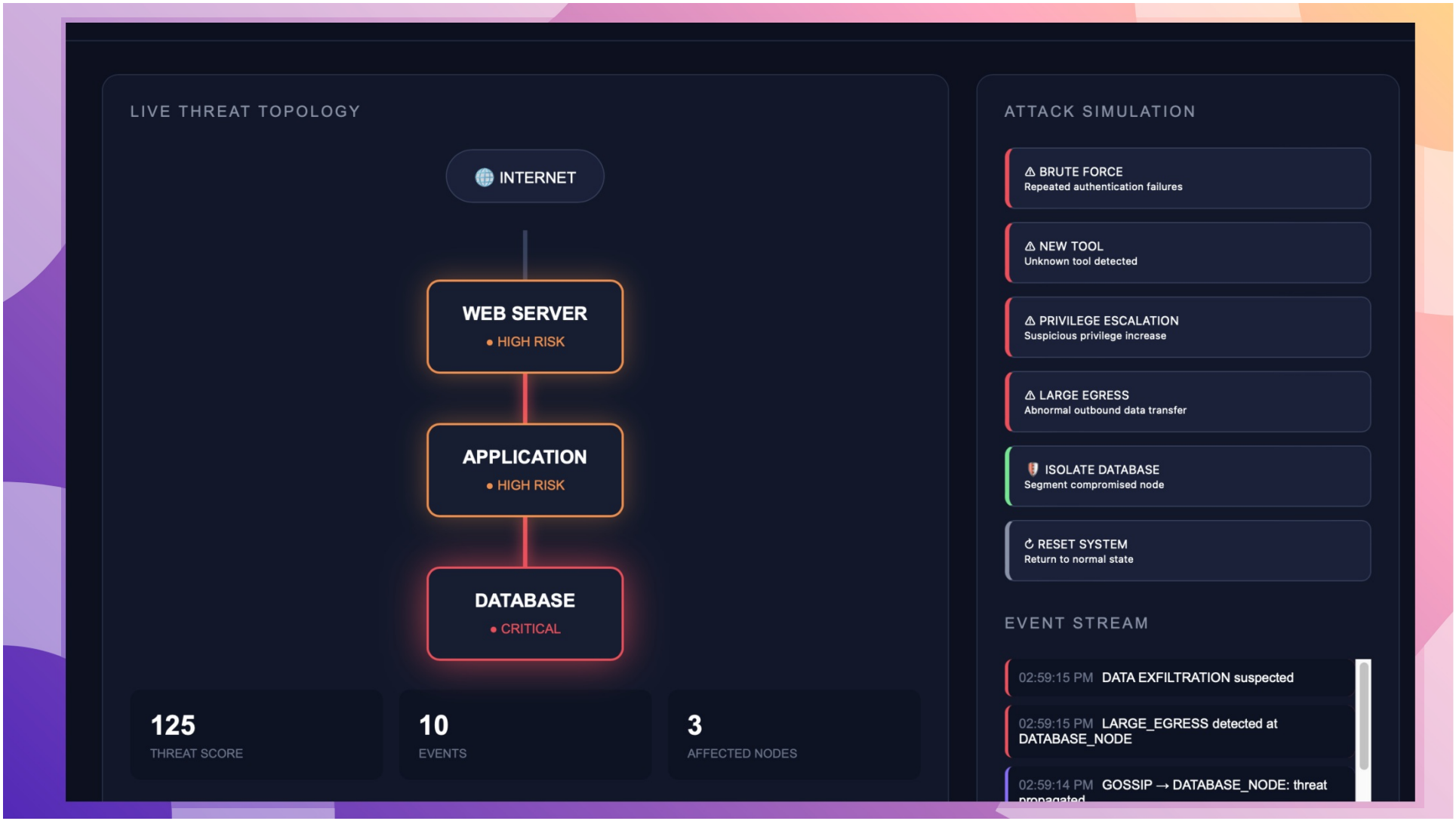
It also demonstrates how relatively simple electronic components can be combined to create a programmable and expandable system.

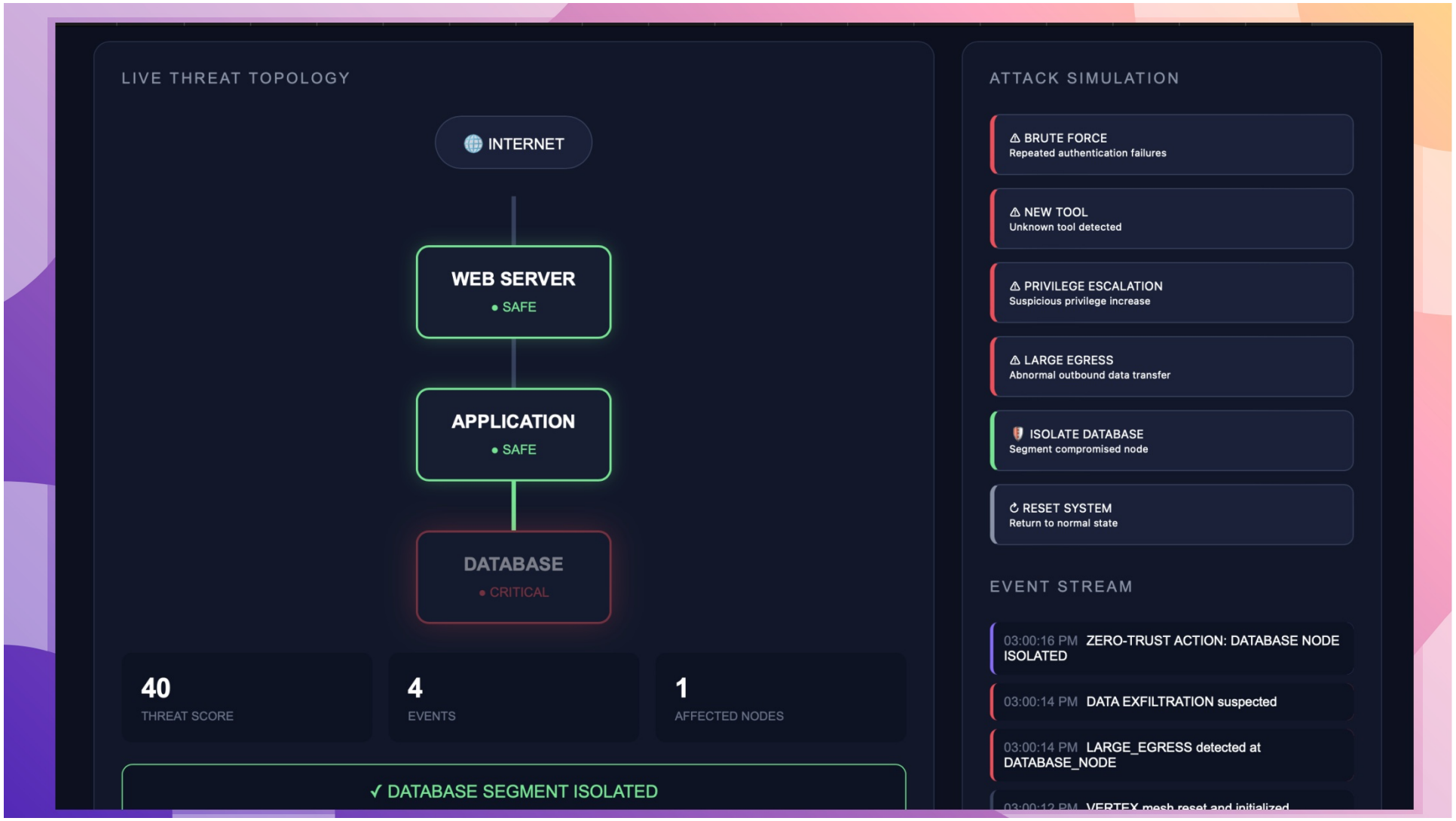

LIVE THREAT TOPOLOGY
INTERNET
WEB SERVER
SAFE
APPLICATION
SAFE
DATABASE
SAFE
0
THREAT SCORE
1
EVENTS
0
AFFECTED NODES
ATTACK SIMULATION
BRUTE FORCE
Repeated authentication failures
NEW TOOL
Unknown tool detected
PRIVILEGE ESCALATION
Suspicious privilege increase
LARGE EGRESS
Abnormal outbound data transfer
ISOLATE DATABASE
Segment compromised node
RESET SYSTEM
Return to normal state
EVENT STREAM
02:52:28 PM VERTEX mesh reset and initialized.

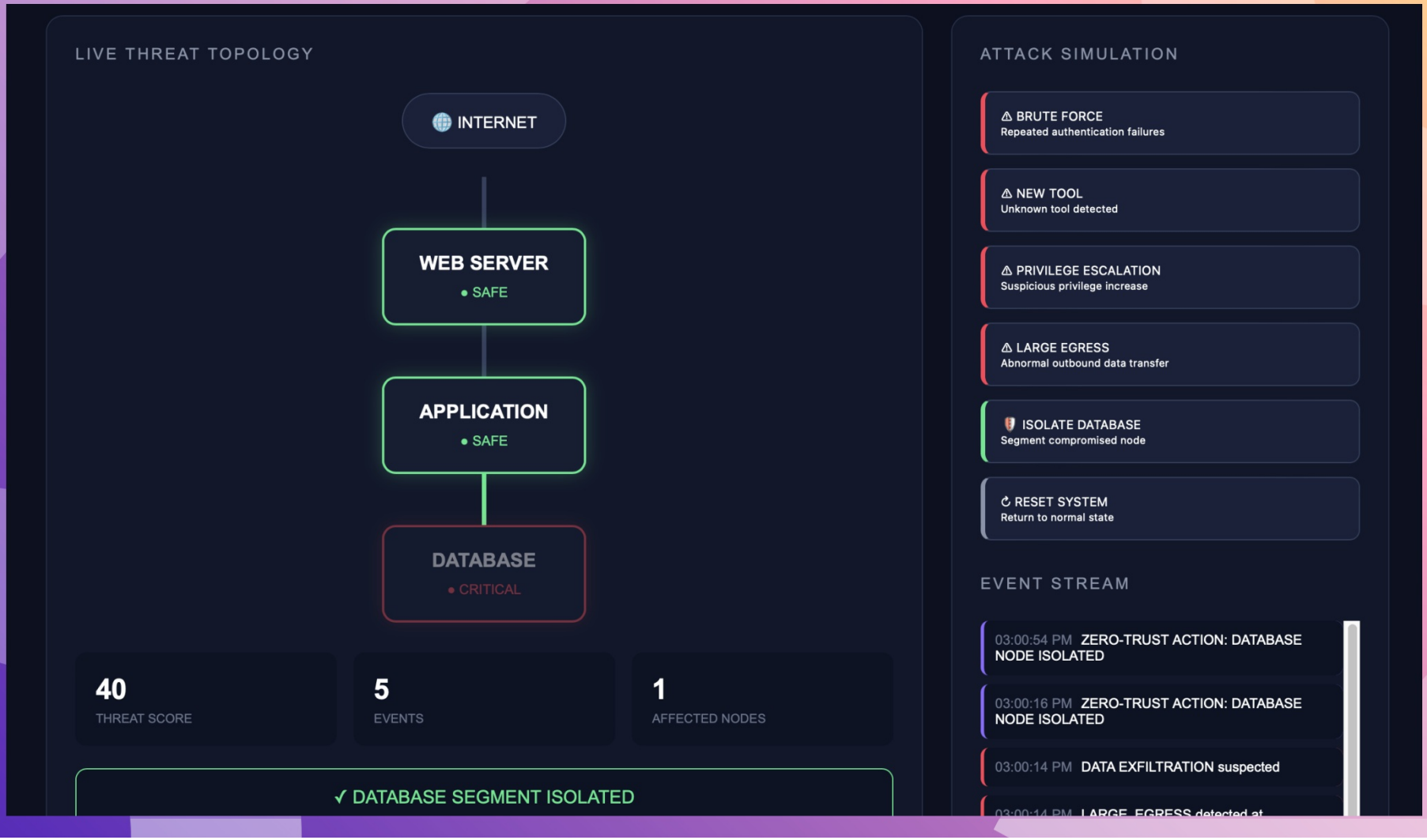

LIVE THREAT TOPOLOGY
INTERNET
WEB SERVER
• SUSPICIOUS
APPLICATION
• SAFE
DATABASE
• SAFE
15
THREAT SCORE
2
EVENTS
1
AFFECTED NODES
ATTACK SIMULATION
⚠ BRUTE FORCE
Repeated authentication failures
⚠ NEW TOOL
Unknown tool detected
⚠ PRIVILEGE ESCALATION
Suspicious privilege increase
⚠ LARGE EGRESS
Abnormal outbound data transfer
ISOLATE DATABASE
Segment compromised node
RESET SYSTEM
Return to normal state
EVENT STREAM
02:56:11 PM BRUTE_FORCE detected at WEB_NODE
02:52:28 PM VERTEX mesh reset and initialized.


LIVE THREAT TOPOLOGY
INTERNET
WEB SERVER
• HIGH RISK
APPLICATION
• ELEVATED
DATABASE
• SAFE
35
THREAT SCORE
4
EVENTS
2
AFFECTED NODES
ATTACK SIMULATION
⚠ BRUTE FORCE
Repeated authentication failures
⚠ NEW TOOL
Unknown tool detected
⚠ PRIVILEGE ESCALATION
Suspicious privilege increase
⚠ LARGE EGRESS
Abnormal outbound data transfer
ISOLATE DATABASE
Segment compromised node
RESET SYSTEM
Return to normal state
EVENT STREAM
02:57:15 PM GOSSIP → APP_NODE: anomaly propagated
02:57:14 PM NEW_TOOL detected at WEB_NODE
02:56:11 PM BRUTE_FORCE detected at WEB_NODE


LIVE THREAT TOPOLOGY
INTERNET
WEB SERVER
• HIGH RISK
APPLICATION
• HIGH RISK
DATABASE
• ELEVATED
60
THREAT SCORE
6
EVENTS
3
AFFECTED NODES
ATTACK SIMULATION
⚠ BRUTE FORCE
Repeated authentication failures
⚠ NEW TOOL
Unknown tool detected
⚠ PRIVILEGE ESCALATION
Suspicious privilege increase
⚠ LARGE EGRESS
Abnormal outbound data transfer
ISOLATE DATABASE
Segment compromised node
RESET SYSTEM
Return to normal state
EVENT STREAM
02:58:16 PM GOSSIP → DATABASE_NODE: threat propagated
02:58:15 PM PRIVILEGE_ESCALATION detected at APP_NODE


LIVE THREAT TOPOLOGY
INTERNET
WEB SERVER
• HIGH RISK
APPLICATION
• HIGH RISK
DATABASE
• CRITICAL
125
THREAT SCORE
10
EVENTS
3
AFFECTED NODES
ATTACK SIMULATION
⚠ BRUTE FORCE
Repeated authentication failures
⚠ NEW TOOL
Unknown tool detected
⚠ PRIVILEGE ESCALATION
Suspicious privilege increase
⚠ LARGE EGRESS
Abnormal outbound data transfer
ISOLATE DATABASE
Segment compromised node
↻ RESET SYSTEM
Return to normal state
EVENT STREAM
02:59:15 PM DATA EXFILTRATION suspected
02:59:15 PM LARGE_EGRESS detected at DATABASE_NODE
02:59:14 PM GOSSIP → DATABASE_NODE: threat


LIVE THREAT TOPOLOGY
INTERNET
WEB SERVER
• SAFE
APPLICATION
• SAFE
DATABASE
• CRITICAL
40
THREAT SCORE
4
EVENTS
1
AFFECTED NODES
✓ DATABASE SEGMENT ISOLATED
ATTACK SIMULATION
⚠ BRUTE FORCE
Repeated authentication failures
⚠ NEW TOOL
Unknown tool detected
⚠ PRIVILEGE ESCALATION
Suspicious privilege increase
⚠ LARGE EGRESS
Abnormal outbound data transfer
ISOLATE DATABASE
Segment compromised node
↻ RESET SYSTEM
Return to normal state
EVENT STREAM
03:00:16 PM ZERO-TRUST ACTION: DATABASE NODE ISOLATED
03:00:14 PM DATA EXFILTRATION suspected
03:00:14 PM LARGE_EGRESS detected at DATABASE_NODE


LIVE THREAT TOPOLOGY
INTERNET
WEB SERVER
• SAFE
APPLICATION
• SAFE
DATABASE
• CRITICAL
40
THREAT SCORE
5
EVENTS
1
AFFECTED NODES
✓ DATABASE SEGMENT ISOLATED
ATTACK SIMULATION
⚠ BRUTE FORCE
Repeated authentication failures
⚠ NEW TOOL
Unknown tool detected
⚠ PRIVILEGE ESCALATION
Suspicious privilege increase
⚠ LARGE EGRESS
Abnormal outbound data transfer
ISOLATE DATABASE
Segment compromised node
↻ RESET SYSTEM
Return to normal state
EVENT STREAM
03:00:54 PM ZERO-TRUST ACTION: DATABASE NODE ISOLATED
03:00:16 PM ZERO-TRUST ACTION: DATABASE NODE ISOLATED
03:00:14 PM DATA EXFILTRATION suspected

